FBISE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 1HA-I 2209 — D(7091) — Section-A only (subjective not in source scan)

Retyped faithfully from the original board paper — same questions, no changes.

Note: CHEMISTRY HSSC-I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 7091. Fill the relevant bubble against each question:

Q1. No. of atoms of oxygen present in 500g of $KClO_3$ is (M.Wt, K=39, Cl=35.5, O=16)

(A) 7.3×10^{24}
(B) 3.61×10^{24}
(C) 2.45×10^{24}
(D) 36.8×10^{23}

Q2. Magnesium reacts with Sulphur to produce MgS. How many grams of MgS can be made from 24g of Mg with excess of Sulphur by the reaction Mg + S → MgS (M.Wt Mg=24, S=32)

(A) 35g
(B) 56g
(C) 58.3g
(D) 12g

Q3. Which of the following is NOT a state function?

(A) Pressure
(B) Volume
(C) Temperature
(D) Work

Q4. If frequency of Photon is 4×10^{15} Hz, then its energy will be: ($h = 6.626\times10^{-34}$ Js)

(A) 26.5×10^{-18} J
(B) 26.5×10^{-19} J
(C) 26.5×10^{19} J
(D) 26.5×10^{18} J

Q5. According to Moseley's law, when metals having large atomic number will be used as an anode then it will produce X-rays of:

(A) Longer wavelength
(B) Low wave number
(C) Shorter wavelength
(D) Low energy

Q6. Total No. of Sigma and Pi bonds in Ethyne CH≡CH are:

(A) 2 Sigma and 3 Pi bonds
(B) 3 Sigma and 2 Pi bonds
(C) 5 Sigma and 3 Pi bonds
(D) 3 Sigma and 5 Pi bonds

Q7. Identify the species with maximum bond angle:

(A) CH_4
(B) NH_3
(C) H_2O
(D) H_2S

Q8. In lead storage battery electrolyte is:

(A) 30% HCl
(B) 30% HNO_3
(C) 30% H_2SO_4
(D) 30% HBr

Q9. According to Le-chatelier's Principle, in an exothermic reaction, what should be done to temperature of system to obtain maximum product at equilibrium?

(A) Kept Low
(B) Kept High
(C) Kept Same
(D) Temperature has no effect

Q10. According to Graham's law which one has highest rate of diffusion? (M. Wt, S=32, O=16, H=1)

(A) SO_2
(B) SO_3
(C) H_2S
(D) SCl_2

Q11. Electric current can pass through graphite in one direction but not through other direction of crystal, such a property is called:

(A) Allotropy
(B) Anisotropy
(C) Isotropy
(D) An isomorphism

Q12. Which one is conductor but not mallable?

(A) Iron
(B) Graphite
(C) Silver
(D) Copper

Q13. Which of the following Concentration Units does NOT depends on Temperature?

(A) Molarity
(B) Molality
(C) $\%^{v}/_{v}$
(D) $\%^{m}/_{v}$

Q14. Which of the following will contain lowest vapor pressure at 50°C?

(A) CH_4
(B) HF
(C) NH_3
(D) H_2O

Q15. A solution of HCl having pH = 4 will be:

(A) 0.4M
(B) 4.0M
(C) 0.0001M
(D) 0.0004M

Q16. Kc = 0.04 at 723K for following reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. Kp of the reaction will be: where ($R = 0.082$ atm dm^3 mol^{-1} K^{-1})

(A) 0.04

(B) 2.37

(C) 0.64

(D) 0.40

Q17. In a mixture containing 44g each of CO_2, SO_2, SO_3 and Cl_2 gases, which of the following will have highest partial pressure according to Dalton's Law? (where molar masses CO_2=44, SO_2=64, SO_3=80, Cl_2=71)

(A) CO_2

(B) SO_2

(C) SO_3

(D) Cl_2

> Note: SUPPLEMENTARY TABLE (Atomic No / Symbol / Mass No): 1 H 1; 2 He 4; 3 Li 7; 4 Be 9; 5 B 11; 6 C 12; 7 N 14; 8 O 16; 9 F 19; 10 Ne 20; 11 Na 23; 12 Mg 24; 13 Al 27; 14 Si 28; 15 P 31; 16 S 32; 17 Cl 35.5; 18 Ar 40; 19 K 39; 20 Ca 40. —1HA-I 2209-7091 (HA)—. Page 1 of 2. Page 2 of 2. Note: the source PDF contains Section-A only; the subjective paper was not present in the scan.